theoretical physics: he found that the mere existence of certain things, even if they are beyond explicit measurement or determination, does make a difference—a difference that can be checked experimentally. Bell showed that if EPR were correct, the results found by two widely separated detectors measuring certain particle properties (spin about various randomly chosen axes, in the approach we have taken) would have to agree more than 50 percent of the time.

Bell had this insight in 1964, but at that time the technology did not exist to undertake the required experiments. By the early 1970s it did. Beginning with Stuart Freedman and John Clauser at Berkeley, followed by Edward Fry and Randall Thompson at Texas A&M, and culminating in the early 1980s with the work of Alain Aspect and collaborators working in France, ever more refined and impressive versions of these experiments were carried out. In the Aspect experiment, for example, the two detectors were placed 13 meters apart and a container of energetic calcium atoms was placed midway between them. Well-understood physics shows that each calcium atom, as it returns to its normal, less energetic state, will emit two photons, traveling back to back, whose spins are perfectly correlated, just as in the example of correlated electron spins we have been discussing. Indeed, in Aspect's experiment, whenever the detector settings are the same, the two photons are measured to have spins that are perfectly aligned. If lights were hooked up to Aspect's detectors to flash red in response to a clockwise spin and blue in response to a counterclockwise spin, the incoming photons would cause the detectors to flash the same color.

But, and this is the crucial point, when Aspect examined data from a large number of runs of the experiment—data in which the left and right detector settings were not always the same but, rather, were randomly and independently varied from run to run—he found that *the detectors did not agree more than 50 percent of the time.*

This is an earth-shattering result. This is the kind of result that should take your breath away. But just in case it hasn't, let me explain further. Aspect's results show that Einstein, Podolsky, and Rosen were proven by experiment—not by theory, not by pondering, but by nature—to be wrong. And that means there has to be something wrong with the reasoning EPR used to conclude that particles possess definite values for